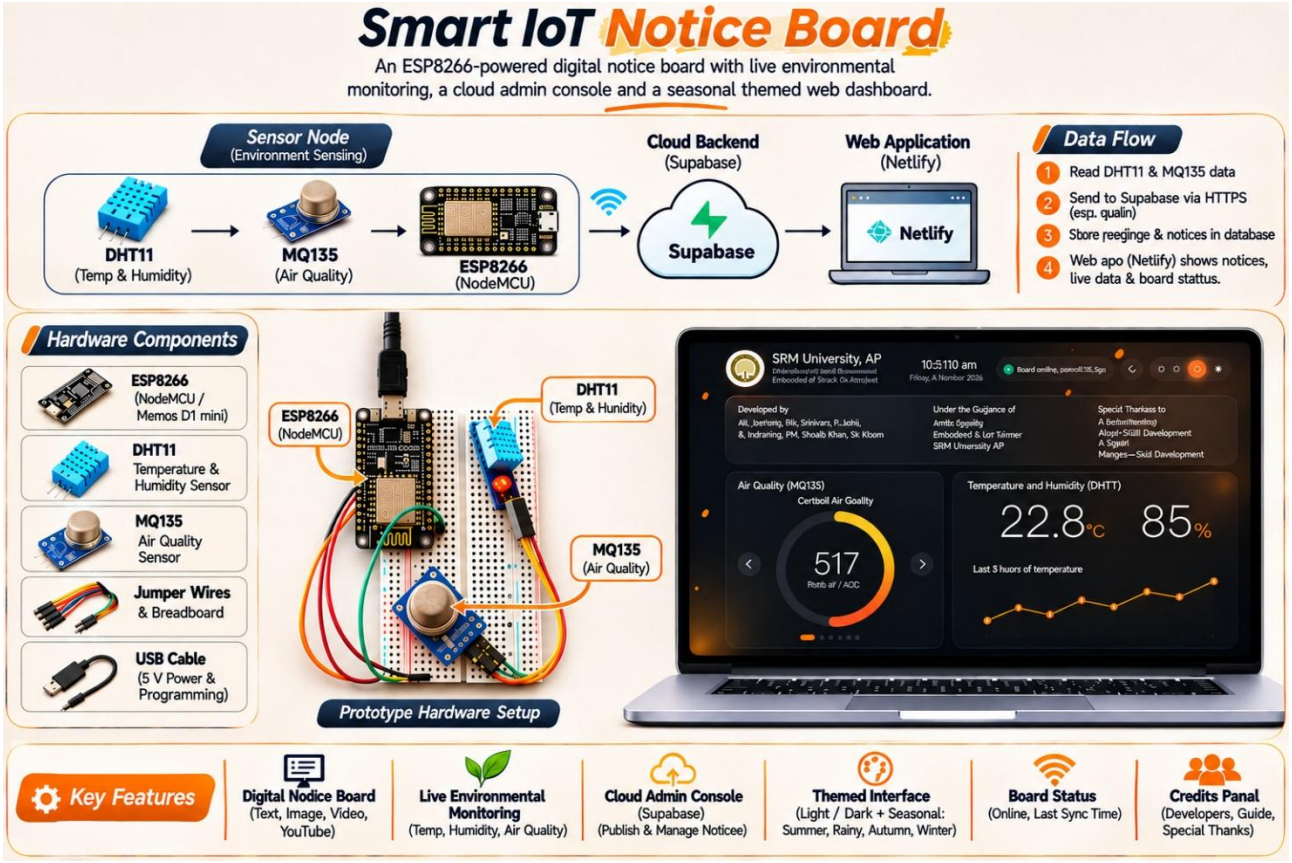
PROJECT DOCUMENTATION

Smart IoT Notice Board

An ESP8266-powered digital notice board with live environmental monitoring, a cloud admin console and a seasonal themed web dashboard.



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1. Abstract

The SRM AP Smart IoT Notice Board is a web-based digital notice board that is linked to a physical ESP8266 sensor node. The node measures temperature and humidity with a DHT11 sensor and air quality with an MQ135 sensor, then sends a reading to a Supabase backend over HTTPS every 10 seconds.

A single-file web application, hosted on Netlify, shows the notices that an administrator publishes and a live environmental dashboard built from the sensor data. The interface supports dark and light modes and four seasonal themes (summer, rainy, autumn and winter), each with its own animated background. This document describes the system design, hardware, firmware, web interface and the results observed during testing.

2. Introduction

2.1 Problem statement

Paper notice boards are slow to update, cannot be seen remotely and say nothing about the environment in which they are placed. A digital board that an administrator can update from any browser, and that also reports live conditions from its own location, solves both problems at low cost.

2.2 Objectives

- Publish notices with formatted text, images, uploaded video or YouTube links from a browser-based admin console.
- Show live temperature, humidity and air-quality readings from an ESP8266 sensor node.
- Use low-cost hardware (ESP8266, DHT11, MQ135) and a managed cloud backend.
- Tie the live features of the site to the sensor board, so the site shows whether the board is online and when it last synced.
- Provide a modern interface with dark and light modes and four animated seasonal themes.

3. System Overview

Feature	Description
Public notice board	Notices appear as cards with a category badge, a date and time, a title, formatted text and optional image or video. A category dropdown (All, General and others) filters the list.
Admin console	A login-protected page for publishing notices. It has category and title fields, a rich-text editor, image upload, video upload, a YouTube link field and Publish / Clear all buttons.
Environmental dashboard	Temperature and humidity from the DHT11 with a last-3-hours temperature trend, plus an MQ135 air-quality card that cycles through nine views.
Board status	A status pill reads "Board online, synced Ns ago", based on the heartbeat the ESP8266 sends.
Live clock	The header shows the current time and date next to the SRM University, AP identity.
Themes	A dark / light toggle and a season selector (summer, rainy, autumn, winter) with a different animation for each season.
Credits panel	Lists the developers, the guide and the special thanks on the dashboard.

4. System Architecture

The system has three parts: a sensor node, a cloud backend and a web application. The ESP8266 pushes readings to the backend, and the web application reads them from there together with the notices.

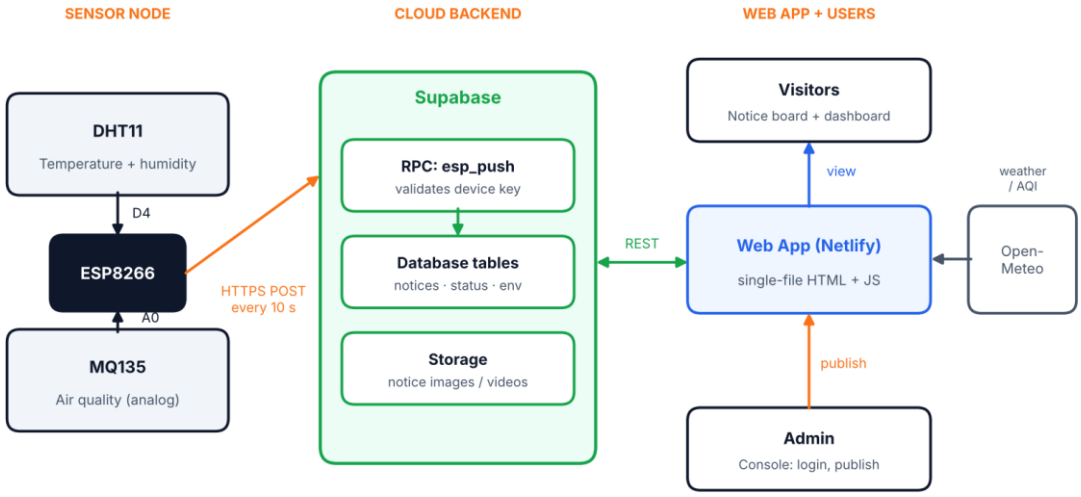


Figure 1: System architecture

4.1 Data flow

1. The DHT11 and MQ135 sensors are read by the ESP8266.
2. Every 10 seconds the firmware builds a JSON payload and sends it with an HTTPS POST to the Supabase RPC function esp_push.
3. The function checks the device key and stores the reading in the database.
4. The web application, hosted on Netlify, reads the latest reading and the notices from Supabase and draws the dashboard.
5. The administrator signs in to the console, publishes notices and uploads media, which are stored in Supabase.

4.2 Technology stack

Layer	Technology
Microcontroller	ESP8266 (NodeMCU or Wemos D1 mini), programmed in the Arduino IDE
Sensors	DHT11 (temperature and humidity), MQ135 (air quality, analog)
Transport	Wi-Fi, HTTPS POST, JSON payload
Backend	Supabase: database tables for notices, status and environment readings, file storage for media, and the esp_push RPC function
Frontend	Single-file HTML, CSS and JavaScript
Hosting	Netlify
External data	Open-Meteo for online weather and air-quality data

5. Hardware

5.1 Components

Component	Role
NodeMCU ESP8266 / Wemos D1 mini	Wi-Fi microcontroller with 3.3 V logic and one analog input (A0)
DHT11 sensor	Digital temperature and humidity sensor
MQ135 gas sensor module	Analog air-quality sensor, needs about 2 minutes to warm up
Resistors (10 kΩ + 20 kΩ)	Voltage divider for A0, only if the MQ135 module output can exceed 3.3 V
USB cable and 5 V supply	Power and programming
Breadboard and jumper wires	Connections

5.2 Pin connections

Sensor	Sensor pin	ESP8266 pin	Notes
DHT11	Data	D4	Defined as DHTPIN in the firmware
MQ135	Analog out (AO)	A0	Read with analogRead, values 0 to 1023
Both	GND	GND	Common ground

6. Firmware

6.1 Board and libraries

- **Board:** ESP8266 core by ESP8266 Community (NodeMCU 1.0 or Wemos D1 mini)
- **Library Manager:** DHT sensor library and Adafruit Unified Sensor
- **Built in:** ESP8266WiFi, ESP8266HTTPClient, WiFiClientSecureBearSSL
- **Serial Monitor:** 115200 baud

6.2 Configuration constants

Constant	Purpose
WIFI_SSID, WIFI_PASS	Wi-Fi network name and password
SB_HOST	Supabase project host, without https://
SB_ANON	Supabase anon (public client) key, sent in the apikey and Authorization headers
DEVICE_KEY	Secret that the esp_push function checks; it must match the value in the Supabase SQL
DHTPIN / DHTTYPE	D4 and DHT11
PUSH_EVERY	10000 ms, the heartbeat interval

6.3 Program flow

1. **setup():** starts the serial port at 115200 baud, prints the reset reason, starts the DHT sensor, disables Wi-Fi sleep, enables auto-reconnect, sets TLS to insecure mode with session caching, builds the endpoint URL and connects to Wi-Fi.
2. **loop():** makes sure Wi-Fi is connected and calls push() once every 10 seconds.
3. **push():** reads temperature and humidity, takes the average of 8 analog readings from the MQ135 (5 ms apart), builds the JSON body and sends it with an HTTPS POST.
4. **Error handling:** if the POST returns a negative code, the firmware prints the error, closes the TLS connection and retries after 3 seconds instead of waiting the full interval.

6.4 Data payload

Each push is a POST to `/rest/v1/rpc/esp_push`. HTTP 200 or 204 means the reading was accepted.

Field	Type	Description
device_key	string	Device secret checked by the RPC function
t	number or null	Temperature in °C, one decimal place
h	number or null	Relative humidity in %, whole number
g	integer	MQ135 raw ADC value (0 to 1023), average of 8 samples
rs	integer	Wi-Fi signal strength (RSSI) in dBm
up	integer	Uptime of the board in seconds

6.5 Reliability design

- **Connection reuse:** one HTTPClient and one TLS session are kept alive, so reconnects skip the heavy TLS handshake on the small ESP8266.
- **Averaging:** eight MQ135 samples are averaged to reduce ADC noise.
- **Failed sensor reads:** if the DHT11 returns NaN, the firmware sends null for that field instead of a wrong number.
- **Steady Wi-Fi:** Wi-Fi sleep is turned off and auto-reconnect is on.
- **Diagnostics:** the reset reason and free heap are printed to the Serial Monitor.

7. Web Application

7.1 Public notice board

Visitors see the notice board as a list of cards. Each card shows a category badge, the date and time it was published, the title, the formatted message and any attached image, video or YouTube embed. The dropdown in the top-right corner filters notices by category. The footer carries the university and department name and an Admin Login link.

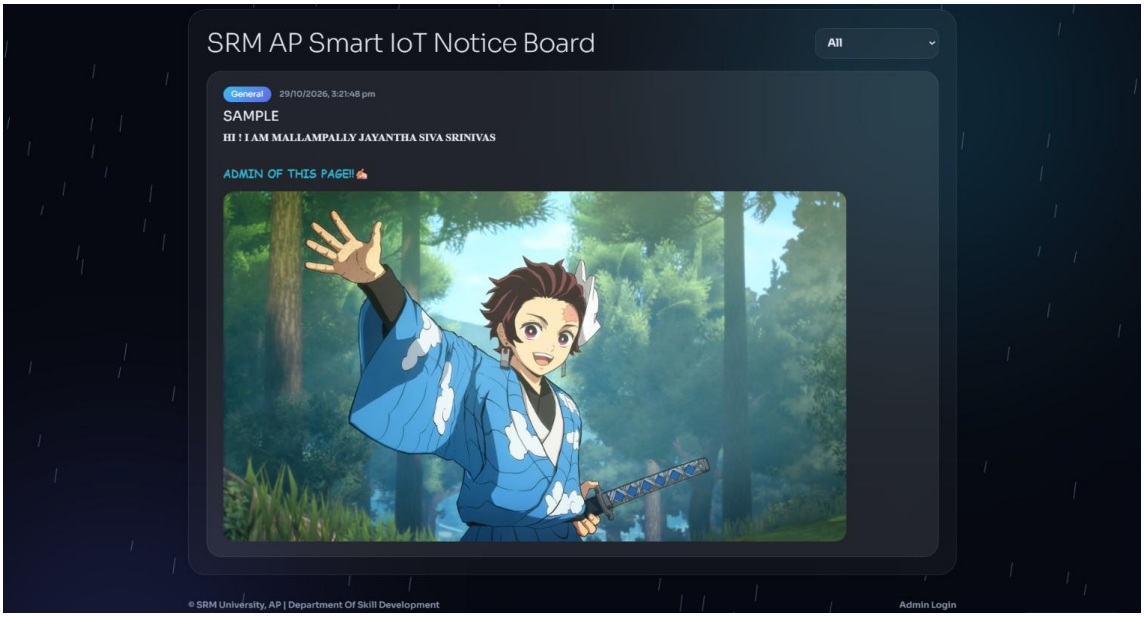


Figure 2: Public notice board showing a sample notice (rainy theme, dark mode)

7.2 Admin console

After logging in, the administrator can publish notices from a single form. The editor lets the author change the font family, size, bold, italic, underline, alignment and text colour, either on selected text or before typing.

Control	Function
Category / Title	Sets the category badge and the notice heading
Information editor	Rich text with font, size, B / I / U, Left / Center / Right and colour picker; Clear text button
Upload image	Attaches an image of any format and resolution; Clear removes it
Upload video	Attaches a video of any format and resolution; Clear removes it
YouTube link	Embeds any YouTube video; Clear link removes it
Publish / Clear all	Publishes the notice, or resets the whole form
Logout	Ends the admin session

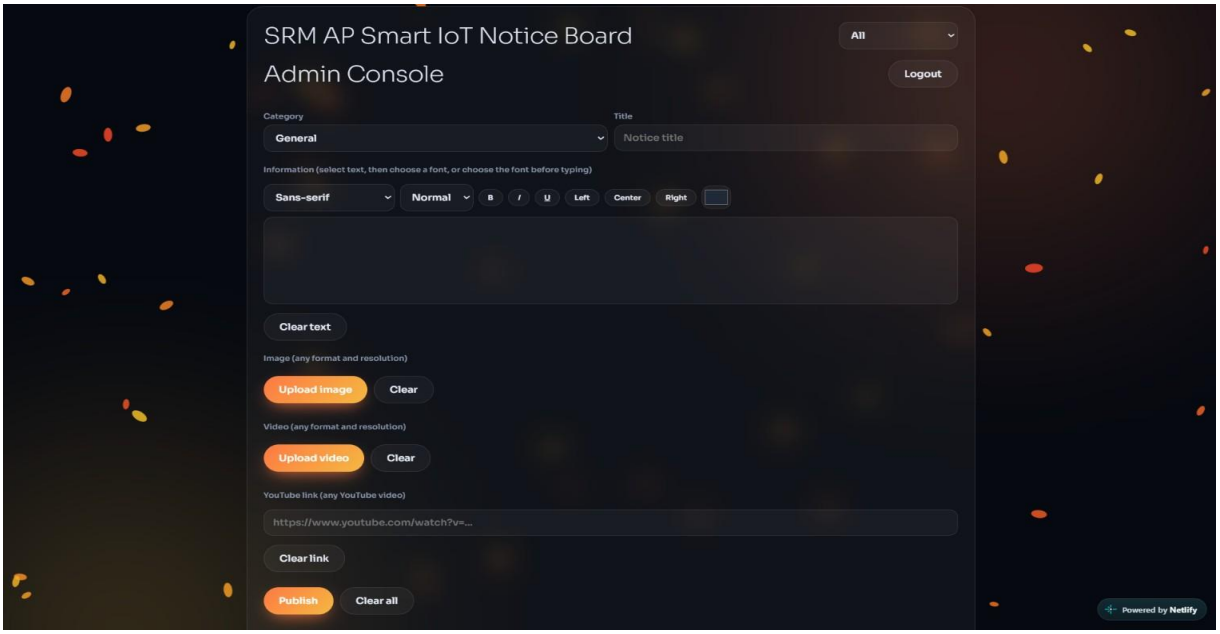


Figure 3: Admin console (autumn theme, dark mode)

7.3 Environmental dashboard

The dashboard header shows the SRM University, AP identity, a live clock and date, the board status pill, the dark / light toggle and the season selector. Below it, a credits panel lists the developers, the guide and the special thanks. The two main cards are the air-quality gauge and the temperature and humidity card.

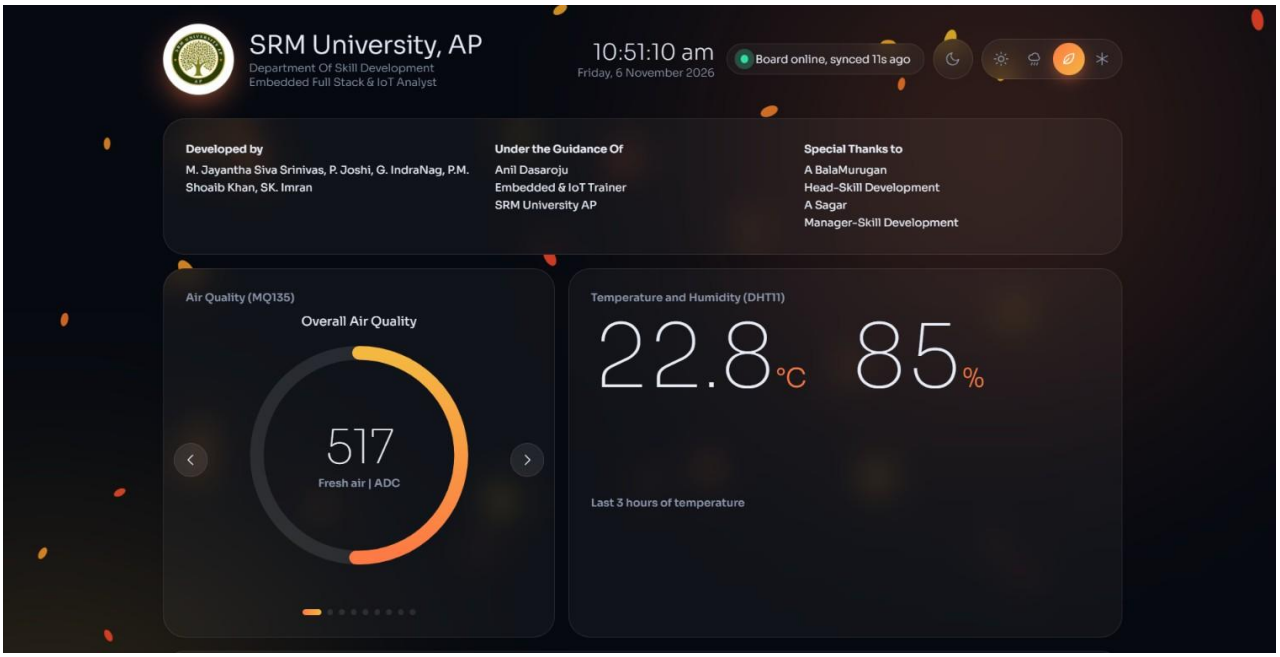


Figure 4: Dashboard overview (autumn theme, dark mode)

Temperature and humidity (DHT11): large live values with a "Last 3 hours of temperature" trend line under them.

Air quality (MQ135): a circular gauge with arrow buttons and nine dots. Each view shows a different gas or indicator estimated from the single MQ135 reading.

#	Gauge view	Unit	Observed value	Status shown
1	Overall Air Quality	ADC	517	Fresh air
2	Ammonia (NH3) / Ammonium	ppm	5.7	Good
3	Carbon Dioxide (CO2)	ppm	501	Good
4	Carbon Monoxide (CO)	ppm	5.2	Good
5	Alcohol / Ethanol	ppm	1.5	Good
6	Benzene and Toluene	ppm	0.6	Good
7	Nitrogen Oxides (NOx)	index	2	Good
8	Smoke / Airborne Particulates	index	0	Good
9	Acetone	ppm	0.4	Good

7.4 Themes and seasons

The interface has two colour modes and four seasonal themes that can be combined freely. The moon / sun button switches between dark and light mode. The four icons beside it (sun, rain cloud, leaf and snowflake) switch the season, which changes the accent colour and the animated background.

Season	Icon	Look and animation
Summer	Sun	Warm orange accents with glowing particles drifting across the background
Rainy	Rain cloud	Blue accents with falling rain streaks
Autumn	Leaf	Orange and red accents with falling leaves
Winter	Snowflake	Cool pale-blue palette with soft drifting snow particles

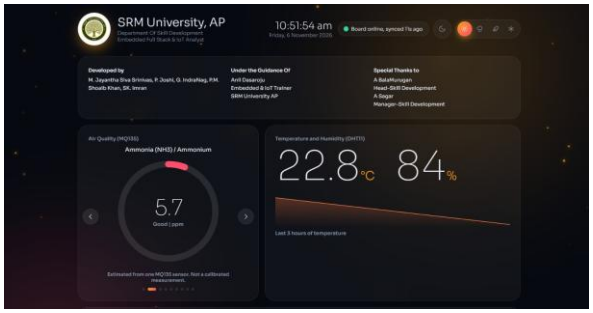


Figure 5: Summer, dark mode (Ammonia view)

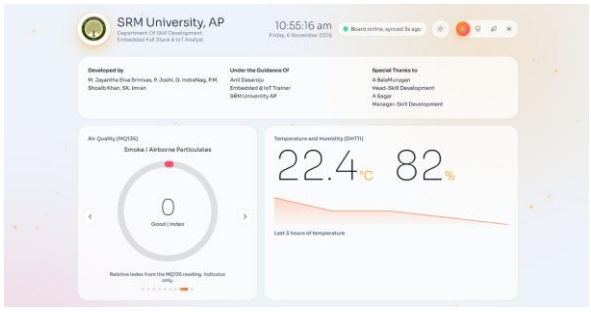


Figure 6: Summer, light mode (Smoke view)

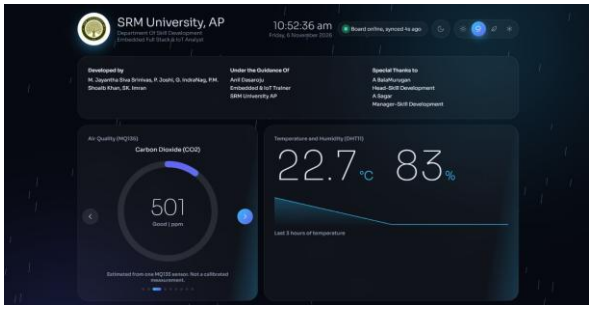


Figure 7: Rainy, dark mode (CO2 view)

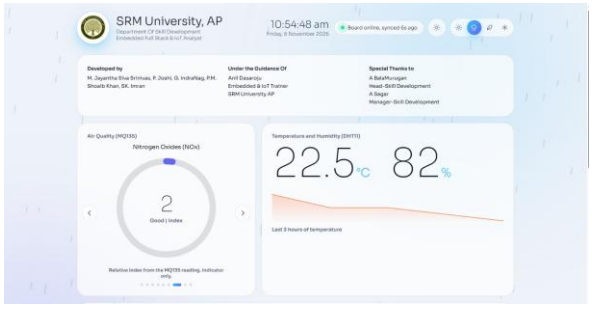


Figure 8: Rainy, light mode (NOx view)

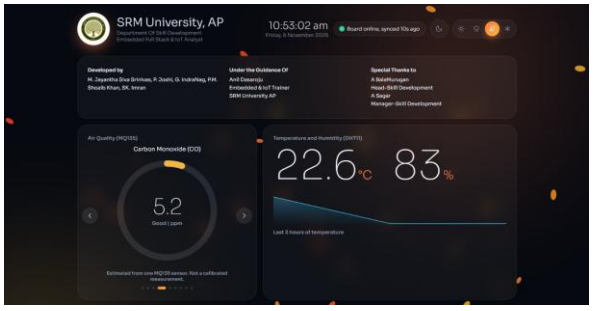


Figure 9: Autumn, dark mode (CO view)

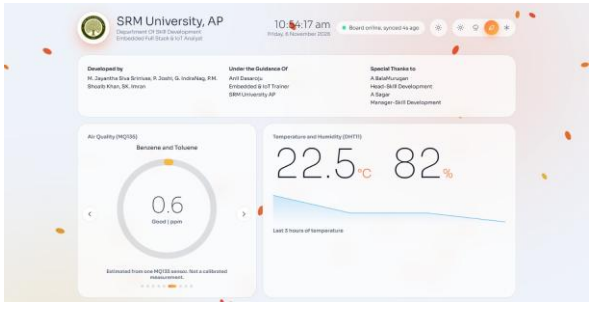


Figure 10: Autumn, light mode (Benzene view)

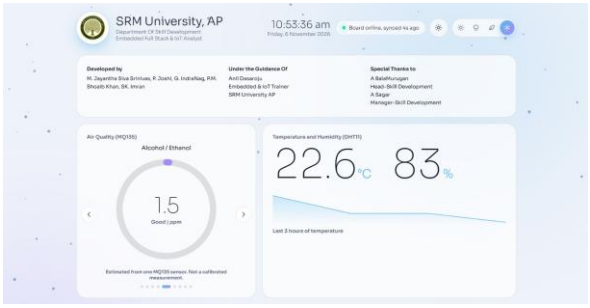


Figure 11: Winter, light mode (Alcohol view)

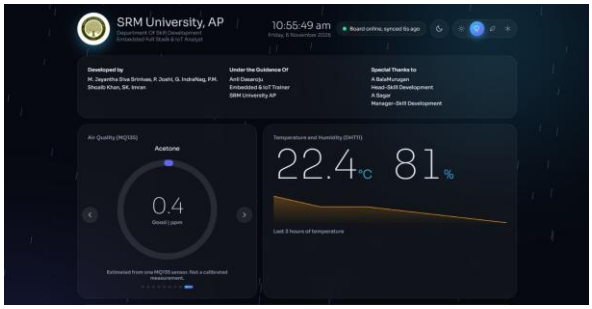


Figure 12: Rainy, dark mode (Acetone view)

7.5 Board status and availability

The status pill in the header reads "Board online, synced Ns ago", where N is the time since the last heartbeat from the ESP8266. By design, the notice board and publishing are available only while the board is online. When the board is off, the site shows only the 10-second introduction page.

8. Results and Observations

The dashboard was captured on Friday, 6 November 2026 between 10:51 and 10:56 am while cycling through the themes and air-quality views.

Time	Mode	Season	Gauge view	Reading	Temp (°C)	Humidity (%)	Board sync
10:51:10	Dark	Autumn	Overall Air Quality	517 ADC, Fresh air	22.8	85	11 s ago
10:51:54	Dark	Summer	Ammonia (NH3)	5.7 ppm, Good	22.8	84	11 s ago
10:52:36	Dark	Rainy	Carbon Dioxide (CO2)	501 ppm, Good	22.7	83	4 s ago
10:53:02	Dark	Autumn	Carbon Monoxide (CO)	5.2 ppm, Good	22.6	83	10 s ago
10:53:36	Light	Winter	Alcohol / Ethanol	1.5 ppm, Good	22.6	83	4 s ago
10:54:17	Light	Autumn	Benzene and Toluene	0.6 ppm, Good	22.5	82	4 s ago
10:54:48	Light	Rainy	Nitrogen Oxides (NOx)	2 index, Good	22.5	82	6 s ago
10:55:16	Light	Summer	Smoke / Airborne Particulates	0 index, Good	22.4	82	3 s ago
10:55:49	Dark	Rainy	Acetone	0.4 ppm, Good	22.4	81	6 s ago

- Temperature drifted from 22.8 °C to 22.4 °C and humidity from 85 % to 81 % over about 4.5 minutes.
- Every air-quality view was rated Good, and the overall view showed Fresh air.
- The board stayed online throughout, with a sync age between 3 and 11 seconds, consistent with the 10-second push interval.
- All four seasons and both colour modes rendered correctly, and the layout stayed the same in every theme.
- The temperature trend line was empty on the first capture and filled in as readings accumulated.
- A sample notice with a category badge, a timestamp, mixed fonts and colours, and an image displayed correctly on the public board.

9. Security and Configuration Notes

- **Device key:** use a long random value and make sure it matches the one stored in the Supabase SQL. Do not leave the sample value from the template.
- **Anon key:** it identifies the project and is meant to be public. Data must be protected by Row Level Security policies and by the device-key check inside the esp_push function.
- **Secrets in source control:** keep the Wi-Fi password and keys out of public repositories and shared documents.
- **TLS:** the firmware calls setInsecure(), which encrypts the connection but does not verify the server. Pin the certificate for a production deployment.
- **Admin access:** use a strong password for the admin login.

10. Limitations

- The MQ135 is a general air-quality sensor and cannot tell individual gases apart. The per-gas values are estimates and need calibration in clean air.
- The DHT11 has limited accuracy (typically about ±2 °C and ±5 % relative humidity).
- The ESP8266 has one analog input, so only one analog sensor can be read without an extra multiplexer.
- The system covers one location with one sensor node.
- Live features depend on Wi-Fi, internet access and the limits of the cloud services used.
- TLS certificate verification is disabled in the current firmware.

11. Future Enhancements

- Replace the DHT11 with a DHT22 or BME280 for better accuracy.
- Pin the server certificate in the firmware.
- Send alerts (for example by Telegram or email) when air quality drops.
- Support several sensor nodes placed in different locations.
- Add over-the-air firmware updates.
- Add notice scheduling and automatic expiry.

12. Conclusion

The SRM AP Smart IoT Notice Board combines a low-cost ESP8266 sensor node, a managed cloud backend and a themed web application into one working system. Notices can be published from a browser, live temperature, humidity and air-quality data is displayed on the same site, and the interface adapts to dark and light modes and four seasonal themes. Testing on 6 November 2026 showed the board syncing steadily and every view and theme rendering correctly.

13. Video and Website Link

Video Link: https://drive.google.com/file/d/1eBH1ALISpCdF8WVlDQ4-lYzuUXHZ0lqw/view?usp=sharing

Website Link: https://jayantha-srmdashboard-1009.netlify.app/